Q2 工业电缆的选择及截面确定
Selection and sizing of industrial cables

先选择适合工业使用条件的电缆结构，再按载流量、压降及功率损耗三项准则确定导体截面。

Select cable construction for industrial service conditions, then size the conductor against current-carrying capacity, voltage drop and power loss.

【1】三类表格的用途 / Purpose of the tables

表格 / Table	用途 / Use
Appendix 7 **安装方式 / Installation methods**	把现场敷设方式对应至参考 Method，确定应查的载流量栏位。 Matches the actual installation to a reference Method, identifying the correct current-rating column.
Tables A5(1)–A5(6) **修正系数 / Correction factors**	按温度〔(1)/(2)〕、成组〔(3)〕、隔热〔(4)〕、保护器件〔(5)〕及封闭地沟〔(6)〕，确定适用于现场条件的所需表列载流量。 Establish the required tabulated capacity for temperature (1/2), grouping (3), thermal insulation (4), protective devices (5) and enclosed trenches (6).
Table A6(1) **载流量及压降 / Ratings and voltage drop**	按截面及 Method 查载流量和压降数据。本表适用于单芯、PVC 绝缘、非铠装铜电缆，基准为 30°C 环境及 70°C 导体；其他电缆应查相应资料。 Provides current ratings and voltage-drop data by size and Method for single-core, PVC-insulated, non-armoured copper cables, at 30°C ambient and 70°C conductor temperature. Other constructions require applicable data.

【2】电缆选型因素 / Factors governing cable selection

因素 / Factor	定性解释 / Qualitative explanation
导体、芯数及绝缘 **Conductor, cores and insulation**	按用途选择铜／铝、芯数及 PVC／XLPE 绝缘；电阻、允许温度及额定数据随结构改变，载流量和压降资料必须匹配。 Select conductor material, core count and PVC/XLPE insulation for the service. Resistance, permissible temperature and ratings vary; use matching current and voltage-drop data.
外部影响、护套及铠装 **External influences, sheath and armour**	工业场所的潮湿、油污、化学品、粉尘及机械损伤风险，决定护套、铠装和防护方式，确保电缆适合现场使用。 Industrial moisture, oil, chemicals, dust and mechanical-damage risks govern sheath, armour and protection, ensuring suitability for the actual environment.
安装方式及排列 **Installation methods and arrangement**	管道、线槽、直接固定、桥架及自由空气的散热不同。按 Appendix 7 确定 Method，并匹配相制、载流导体数量及排列选栏。 Conduit, trunking, clipped-direct, tray and free-air installations cool differently. Use Appendix 7 to identify the Method, matching phase, loaded-conductor count and formation.

【3】导体截面确定因素 / Factors governing conductor sizing

因素 / Factor	定性解释 / Qualitative explanation
负载及设计电流 Ib **Loading and design current**	按设备负载及实际使用情况确定 Ib；载流能力须满足负载，负载增大通常需增大截面，以限制导体温升。 Determine Ib from equipment loading and service conditions. Capacity must meet the load; higher current generally requires a larger cross-section to limit temperature rise.
环境温度 Ca **Ambient temperature**	温度升高使可用载流量降低。一般采用 A5(1)；BS3036 半封闭熔断器承担过载保护时采用 A5(2)，两表不同时使用。 Higher temperature reduces available capacity. Use A5(1) generally, or A5(2) when a BS3036 semi-enclosed fuse provides overload protection; these are alternatives.
电缆成组 Cg **Grouping**	成束或围护内回路互相加热。按 A5(3)的回路数量、间距及排列选 Cg；散热越受限，通常越需增大截面。 Bunched or enclosed circuits heat one another. Select Cg from A5(3) for circuit count, spacing and arrangement; restricted cooling generally requires a larger size.
隔热材料 Ci **Thermal insulation**	隔热材料阻碍散热。按实际安装条件及 A5(4)的适用范围折减；参考 Method 已计入的同一隔热影响，应避免重复修正。 Thermal insulation restricts cooling. Use the actual installation and applicable A5(4) conditions, avoiding duplicate correction for an effect already included in the reference Method.
保护器件额定值 In 及 Cd **Overcurrent protective device**	按 In 及适用系数求所需表列载流量，使电缆与保护配合。A5(5)的 Cd：BS3036 半封闭熔断器为 0.725，其他所列器件为 1.0。 Coordinate cable capacity with In and applicable factors. A5(5) gives Cd = 0.725 for BS3036 semi-enclosed fuses and 1.0 for the other listed devices.
封闭地沟 **Enclosed trenches**	封闭地沟积热影响散热。按沟体、盖板、导体截面、电缆数量及间距选 A5(6)系数，修正相应 Method 12／13 的基础额定值。 Enclosed trenches retain heat. Apply A5(6) for trench construction, cover, conductor size, cable count and spacing to the appropriate Method 12/13 base rating.
长度及压降 **Cable length and voltage drop**	长度和电流越大，压降通常越大。按电缆、截面、回路及排列查压降值，并计入上游压降；即使载流量合格，也可能需增大截面。 Longer routes and higher current increase voltage drop. Use matching cable, size, circuit and formation data, including upstream drop; current-capacity compliance alone may not establish adequate size.
功率损耗及能源效益 **Power loss and energy efficiency**	工业回路长期带载时须考虑 I^2R 损耗。相同电流下，较大截面降低电阻和损耗，改善能源效益。 For long-running industrial circuits, consider I^2R loss. At a given current, a larger cross-section reduces resistance and losses, improving energy efficiency.

$$\mathbf{It = In / (Ca \times Cg \times Ci \times Cd)}$$

过载保护配合时，It 为所需最低表列载流量，选择 Ita ≥ It；按现场条件采用适用系数。

For overload coordination, It is the minimum required tabulated capacity; select Ita ≥ It using the applicable factors.

【4】简短查表示例 / Brief table application

例：三相 380V，Ib = In = 20A（MCB），45℃、40m；PVC 单芯铜缆以表面管道 Method 3 敷设，其他修正系数为 1，上游压降 1%。

Example: 380V three-phase, Ib = In = 20A (MCB), 45°C, 40m; single-core PVC copper cables in surface conduit, Method 3; other factors 1, upstream drop 1%.

A5(1)：Ca = 0.79 ⇒ It = 20 / 0.79 = 25.3A。

A6(1)，Method 3：4mm²，Ita = 28A ≥ 25.3A；VDta,z = 9.5mV/(A·m)。

VD = 9.5 × 20 × 40 / 1000 = 7.6V；7.6 / 380 × 100% = 2%。

总压降 / Total drop = 1% + 2% = 3% < 4%（本例限值 / example limit）。